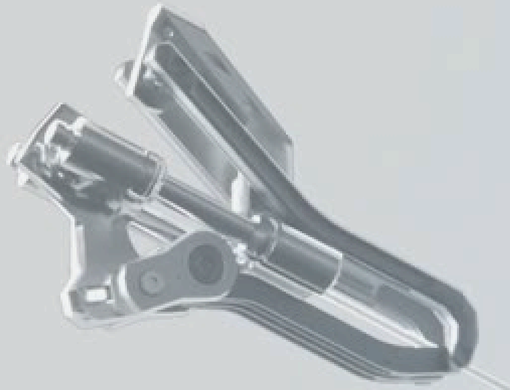
DESIGN PROJECT

MOSHI

DESK ROBOT COMPANION

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2K24/BD/046





Project Overview

MOSHI IS AN EXPRESSIVE ROBOTIC COMPANION DESIGNED TO EXPLORE EMOTIONAL INTERACTION BETWEEN HUMANS AND MACHINES THROUGH MOVEMENT, BEHAVIOR, AND DIGITAL EXPRESSION. THE PROJECT REINTERPRETS AN OPEN-SOURCE QUADRUPED ROBOTIC PLATFORM INTO A DESIGN-LED INTERACTIVE EXPERIENCE FOCUSED ON COMPANIONSHIP, PLAYFULNESS, AND APPROACHABLE TECHNOLOGY.

RATHER THAN TREATING ROBOTICS PURELY AS A MECHANICAL OR ENGINEERING CHALLENGE, MOSHI INVESTIGATES HOW MOVEMENT, PERSONALITY, VISUAL LANGUAGE, AND RESPONSIVE BEHAVIOR CAN CREATE EMOTIONAL ENGAGEMENT IN EVERYDAY ENVIRONMENTS.

THE PROJECT COMBINES:

- INTERACTION DESIGN
- PRODUCT DESIGN
- HUMAN-ROBOT INTERACTION
- PHYSICAL COMPUTING
- EMOTIONAL DESIGN
- EMBEDDED SYSTEMS
- BEHAVIORAL ANIMATION
- RAPID PROTOTYPING

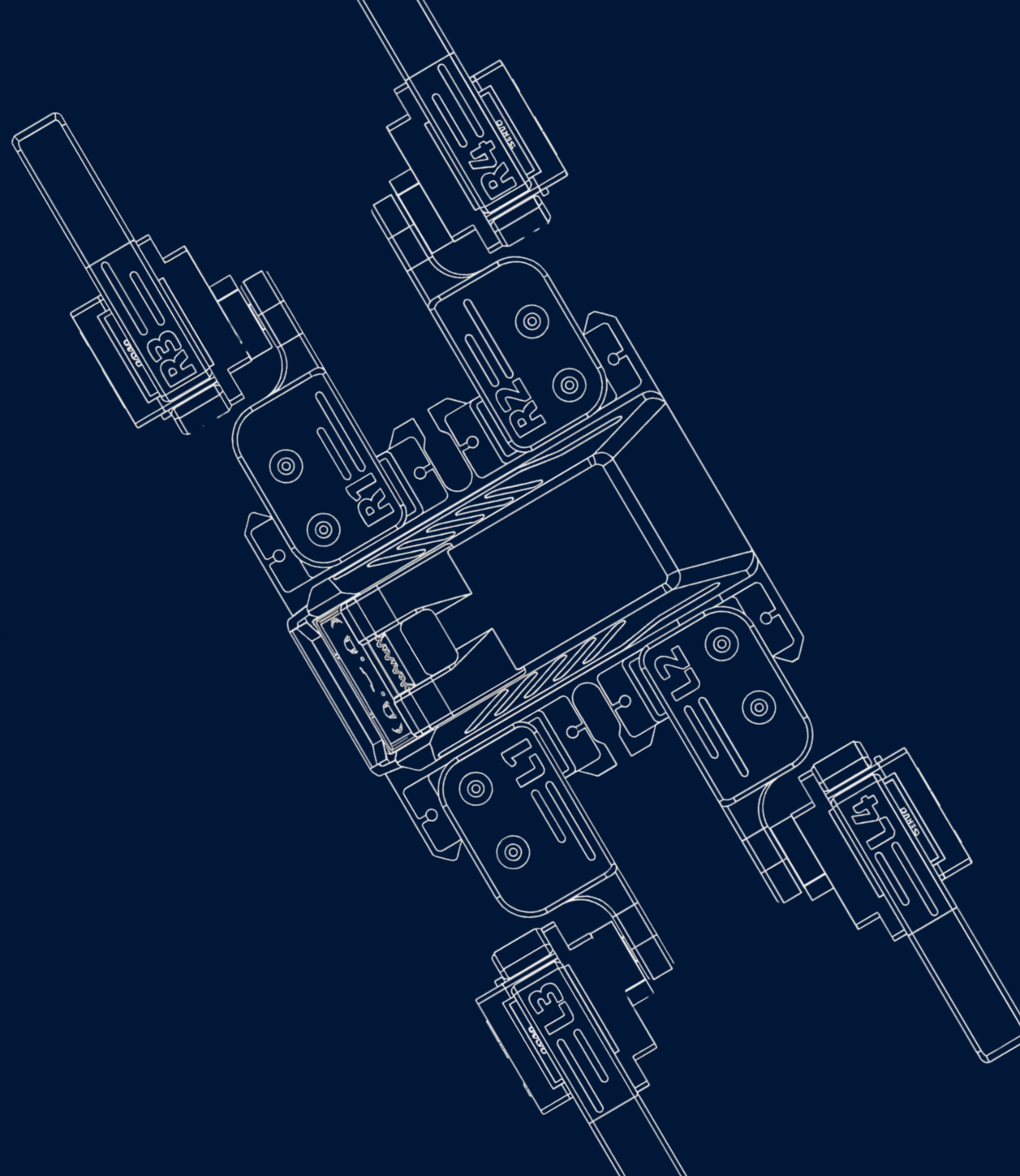
MOSHI WAS DEVELOPED AS A HIGH-LEVEL DESIGN AND TECHNOLOGY EXPLORATION USING THE OPEN-SOURCE SESAME ROBOT FRAMEWORK AS A TECHNICAL FOUNDATION.

Project Context

TODAY, MOST INTERACTIONS WITH TECHNOLOGY HAPPEN THROUGH SCREENS, MAKING DIGITAL EXPERIENCES INCREASINGLY DETACHED FROM PHYSICAL AND EMOTIONAL ENGAGEMENT. AT THE SAME TIME, ROBOTICS TECHNOLOGY CONTINUES TO ADVANCE RAPIDLY, YET MANY ROBOTIC SYSTEMS STILL FEEL HIGHLY TECHNICAL, INDUSTRIAL, AND INTIMIDATING FOR EVERYDAY USERS.

MOST AFFORDABLE ROBOTIC PLATFORMS FOCUS HEAVILY ON ENGINEERING FUNCTIONALITY, LOCOMOTION SYSTEMS, AND TECHNICAL PERFORMANCE WHILE LACKING EMOTIONAL ACCESSIBILITY AND INTERACTIVE PERSONALITY. THIS CREATES A DISCONNECT BETWEEN TECHNOLOGICAL CAPABILITY AND HUMAN ENGAGEMENT.

MOSHI EXPLORES HOW ROBOTICS CAN BECOME MORE EMOTIONALLY APPROACHABLE THROUGH EXPRESSIVE MOVEMENT, BEHAVIORAL INTERACTION, AND PLAYFUL COMMUNICATION SYSTEMS.



Problem Statement

MOST EXISTING ROBOTIC SYSTEMS PRIORITIZE FUNCTIONALITY AND TECHNICAL COMPLEXITY OVER EMOTIONAL INTERACTION AND ACCESSIBILITY. AS A RESULT, MANY ROBOTS FEEL MECHANICAL, INTIMIDATING, OR DISCONNECTED FROM EVERYDAY USERS.

THERE IS AN OPPORTUNITY TO RETHINK ROBOTICS AS EMOTIONALLY ENGAGING COMPANIONS THAT COMMUNICATE THROUGH MOVEMENT, PERSONALITY, AND BEHAVIOR RATHER THAN PURELY UTILITY-DRIVEN MACHINES.

THE PROJECT INVESTIGATES HOW EXPRESSIVE MOVEMENT, VISUAL COMMUNICATION, AND INTERACTION DESIGN CAN TRANSFORM A ROBOTIC PLATFORM INTO A MORE APPROACHABLE AND RELATABLE EXPERIENCE.



How might we design a robotic companion that feels emotionally engaging, communicates through expressive behavior, and combines engineering functionality with human-centered interaction design?

THE GOAL WAS NOT SIMPLY TO BUILD A FUNCTIONING ROBOT, BUT TO EXPLORE HOW ROBOTICS CAN CREATE EMOTIONAL CONNECTION THROUGH MOVEMENT, INTERACTION, AND PERSONALITY.


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Introducing Moshi

OBJECTIVES

THE PRIMARY OBJECTIVE OF MOSHI WAS TO CREATE AN EMOTIONALLY EXPRESSIVE ROBOTIC COMPANION CAPABLE OF ENGAGING USERS THROUGH MOVEMENT, POSTURE, TIMING, AND VISUAL EXPRESSION.

THE PROJECT ALSO AIMED TO EXPLORE HOW INTERACTION DESIGN PRINCIPLES COULD BE INTEGRATED INTO ROBOTICS TO CREATE MORE APPROACHABLE TECHNOLOGICAL EXPERIENCES. FROM A TECHNICAL PERSPECTIVE, THE PROJECT FOCUSED ON DEVELOPING A FUNCTIONAL QUADRUPED LOCOMOTION SYSTEM INTEGRATED WITH EXPRESSIVE OLED-BASED FACIAL ANIMATIONS AND A WIRELESS INTERACTION INTERFACE.

ANOTHER MAJOR OBJECTIVE WAS TO TRANSFORM ROBOTICS FROM A CODING-HEAVY ENGINEERING SYSTEM INTO A MORE ACCESSIBLE INTERACTIVE PRODUCT THROUGH BROWSER-BASED CONTROL AND EMOTIONAL COMMUNICATION SYSTEMS.

Research followed

THE RESEARCH PHASE FOCUSED ON UNDERSTANDING HOW HUMANS EMOTIONALLY INTERPRET ROBOTIC BEHAVIOR AND HOW INTERACTION SYSTEMS INFLUENCE ENGAGEMENT WITH TECHNOLOGY.

RESEARCH AREAS INCLUDED COMPANION ROBOTICS, EMOTIONAL COMPUTING, ANTHROPOMORPHISM, BEHAVIORAL ANIMATION, HUMAN-ROBOT INTERACTION, AND PHYSICAL COMPUTING SYSTEMS. EXISTING ROBOTIC PRODUCTS SUCH AS SONY AIBO, LOVOT, VECTOR, EMO ROBOT, AND PETOI BITTLE WERE STUDIED TO UNDERSTAND HOW MOVEMENT, EXPRESSIONS, AND RESPONSIVENESS INFLUENCE EMOTIONAL PERCEPTION.

ONE OF THE KEY FINDINGS FROM THE RESEARCH WAS THAT HUMANS NATURALLY ASSIGN PERSONALITY AND EMOTION TO EVEN SIMPLE ROBOTIC BEHAVIORS SUCH AS BLINKING, POSTURE SHIFTS, MOVEMENT TIMING, AND IDLE ANIMATIONS. THIS INSIGHT BECAME ONE OF THE FOUNDATIONAL PRINCIPLES BEHIND MOSHI’S INTERACTION SYSTEM.



User Survey

A SMALL USER SURVEY WAS CONDUCTED TO UNDERSTAND HOW USERS EMOTIONALLY PERCEIVE ROBOTIC SYSTEMS AND WHAT QUALITIES MAKE TECHNOLOGY FEEL APPROACHABLE.

THE SURVEY EXPLORED QUESTIONS RELATED TO EMOTIONAL INTERACTION, PREFERRED MOVEMENT STYLES, ROBOTIC PERSONALITY, AND VISUAL COMMUNICATION SYSTEMS. MOST PARTICIPANTS PREFERRED EXPRESSIVE EYES AND SUBTLE MOVEMENT OVER REALISTIC HUMANOID FACES OR AGGRESSIVE MECHANICAL MOTION.

USERS STRONGLY ASSOCIATED BLINKING WITH AWARENESS, HEAD TILTING WITH CURIOSITY, AND BOUNCING MOVEMENT WITH EXCITEMENT. PARTICIPANTS ALSO PREFERRED SMALLER COMPANION-STYLE ROBOTS OVER INDUSTRIAL ROBOTIC AESTHETICS.
THESE FINDINGS DIRECTLY INFLUENCED MOSHI'S BEHAVIORAL SYSTEM, MOVEMENT LANGUAGE, AND VISUAL DESIGN DIRECTION.

Concept Development

THE PROJECT INITIALLY EXPLORED MULTIPLE ROBOTIC ARCHETYPES INCLUDING INDUSTRIAL ROBOTS, FUTURISTIC AI ASSISTANTS, UTILITY ROBOTS, AND PET-INSPIRED COMPANIONS.

HOWEVER, THE FINAL DIRECTION FOCUSED ON CREATING A PLAYFUL COMPANION ROBOT INSPIRED BY CAT-LIKE BEHAVIOR AND MOVEMENT. THIS DIRECTION WAS SELECTED BECAUSE FELINE MOVEMENT NATURALLY COMMUNICATES CURIOSITY, CALMNESS, ATTENTIVENESS, AND EMOTIONAL SUBTLETY THROUGH POSTURE AND BEHAVIORAL TIMING.

THE CONCEPT SHIFTED THE PROJECT AWAY FROM PURELY TECHNICAL ROBOTICS AND TOWARD A CHARACTER-DRIVEN INTERACTION EXPERIENCE.



Why cat iteration?

THE CAT-INSPIRED ITERATION WAS CHOSEN BECAUSE CATS COMMUNICATE EMOTION THROUGH SUBTLE MOVEMENT RATHER THAN EXAGGERATED ACTIONS. THEIR BODY LANGUAGE, BLINKING PATTERNS, POSTURE SHIFTS, AND REACTION TIMING CREATE STRONG EMOTIONAL INTERPRETATION DESPITE MINIMAL MOVEMENT.

THIS ALIGNED CLOSELY WITH THE PROJECT GOAL OF EXPLORING EMOTIONAL INTERACTION THROUGH BEHAVIOR-DRIVEN ROBOTICS.

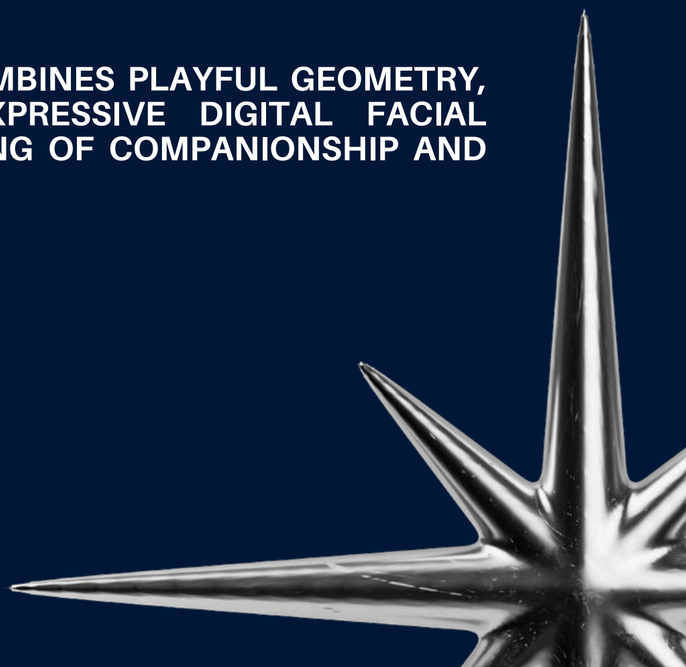
THE CAT-INSPIRED VISUAL IDENTITY ALSO HELPED REDUCE ROBOTIC INTIMIDATION BY MAKING THE SYSTEM FEEL SOFTER, MORE FAMILIAR, AND EMOTIONALLY APPROACHABLE. THE EXPRESSIVE FACE DESIGN STRENGTHENED THE CHARACTER IDENTITY OF MOSHI AND ALLOWED THE ROBOT TO FEEL MORE COMPANION-LIKE RATHER THAN MACHINE-LIKE.

Naming & Identity

THE NAME “MOSHI” WAS SELECTED TO CREATE A SOFT, PLAYFUL, AND EMOTIONALLY WARM IDENTITY FOR THE ROBOT.

THE INTENTION BEHIND THE NAME WAS TO MAKE THE ROBOT FEEL APPROACHABLE AND CHARACTER-DRIVEN RATHER THAN PURELY TECHNICAL. THE NAMING ALSO SUPPORTED THE PROJECT’S LARGER GOAL OF TRANSFORMING ROBOTICS INTO EMOTIONALLY ENGAGING EXPERIENCES.

THE VISUAL IDENTITY OF MOSHI COMBINES PLAYFUL GEOMETRY, COMPACT PROPORTIONS, AND EXPRESSIVE DIGITAL FACIAL SYSTEMS TO REINFORCE THE FEELING OF COMPANIONSHIP AND INTERACTION.



Design Process

THE PROJECT FOLLOWED AN ITERATIVE DESIGN PROCESS COMBINING RESEARCH, FORM EXPLORATION, TECHNICAL DEVELOPMENT, ASSEMBLY, INTERACTION DESIGN, AND BEHAVIORAL REFINEMENT.

THE INITIAL STAGES FOCUSED ON UNDERSTANDING EMOTIONAL ROBOTICS AND MOVEMENT-BASED INTERACTION SYSTEMS. THIS WAS FOLLOWED BY EXTENSIVE FORM EXPLORATION TO DETERMINE THE VISUAL LANGUAGE AND POSTURE SYSTEM OF THE ROBOT.

THE TECHNICAL DEVELOPMENT PHASE FOCUSED ON LOCOMOTION SYSTEMS, EMBEDDED ELECTRONICS, EXPRESSIVE FACE INTEGRATION, AND WIRELESS COMMUNICATION ARCHITECTURE. MULTIPLE ASSEMBLY AND TESTING ITERATIONS WERE CONDUCTED TO IMPROVE MOVEMENT QUALITY, STABILITY, AND INTERACTION SMOOTHNESS.

THE FINAL PHASE FOCUSED ON REFINING BEHAVIORAL INTERACTION AND EXPRESSIVE COMMUNICATION THROUGH SYNCHRONIZED MOVEMENT AND FACIAL ANIMATION SYSTEMS.

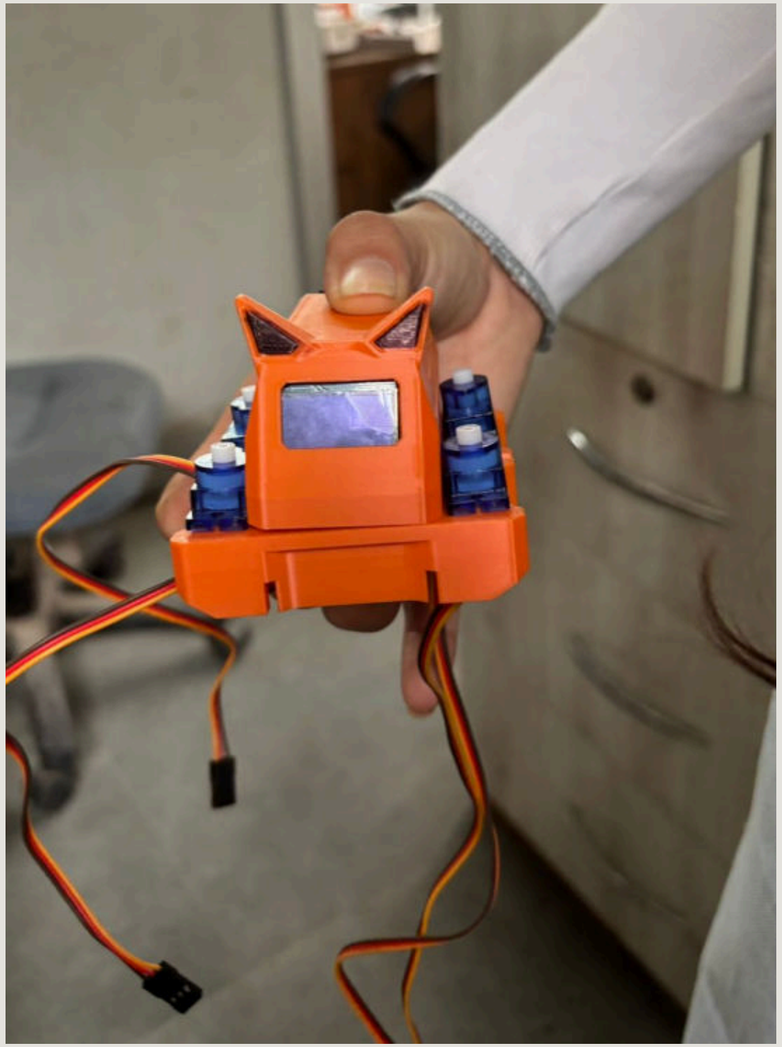
Visual design language

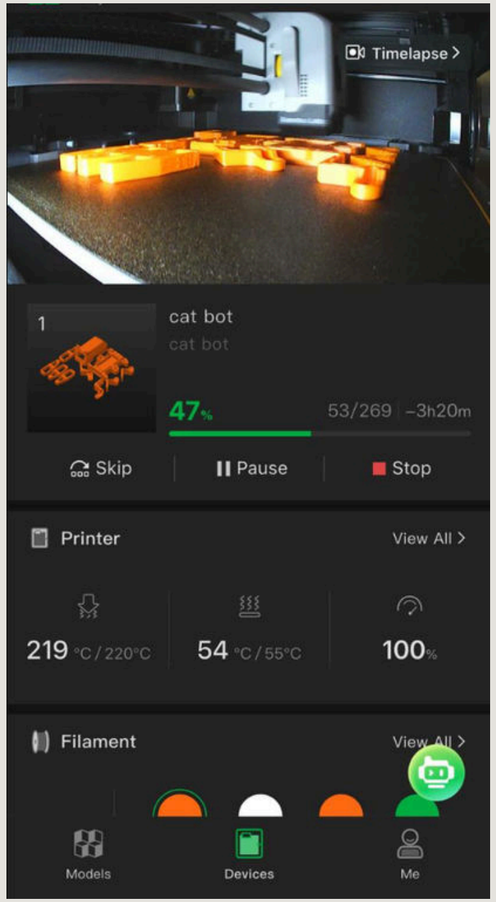
THE VISUAL LANGUAGE OF MOSHI INTENTIONALLY AVOIDS AGGRESSIVE INDUSTRIAL ROBOTICS AESTHETICS. INSTEAD, THE PROJECT USES ROUNDED GEOMETRY, COMPACT PROPORTIONS, PLAYFUL FORMS, AND EXPRESSIVE FACIAL PLACEMENT TO CREATE A SOFTER AND MORE APPROACHABLE ROBOTIC IDENTITY.

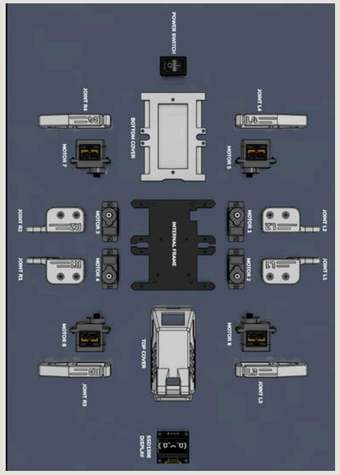
THE DESIGN DRAWS INSPIRATION FROM ANIMATED CHARACTER DESIGN, DESKTOP COMPANION PRODUCTS, AND MINIMAL ROBOTIC PETS. THE GOAL WAS TO CREATE A ROBOT THAT VISUALLY COMMUNICATES FRIENDLINESS AND EMOTIONAL SAFETY.

THE SIMPLIFIED FORM LANGUAGE ALSO HELPED STRENGTHEN THE EMOTIONAL READABILITY OF MOVEMENT AND EXPRESSIONS.

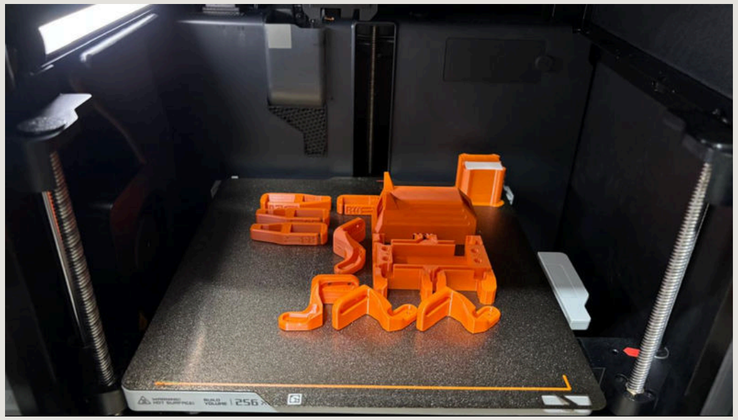


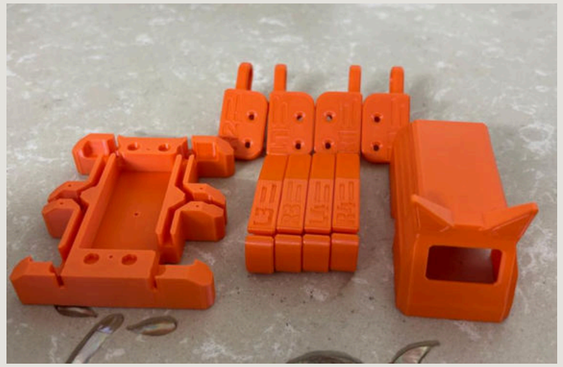


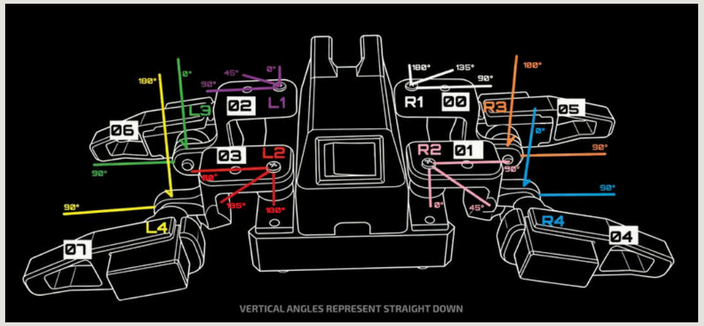




CAD design analysis







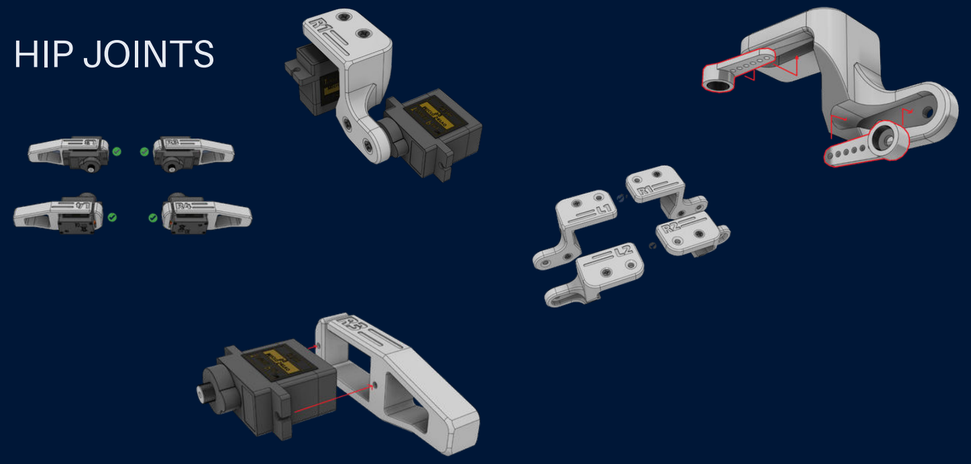
THE CAD FRAMEWORK FROM THE SESAME ROBOT PROJECT WAS SELECTED BECAUSE IT EFFECTIVELY BALANCED MODULARITY, AFFORDABILITY, LIGHTWEIGHT CONSTRUCTION, AND EXPRESSIVE MOVEMENT CAPABILITY.

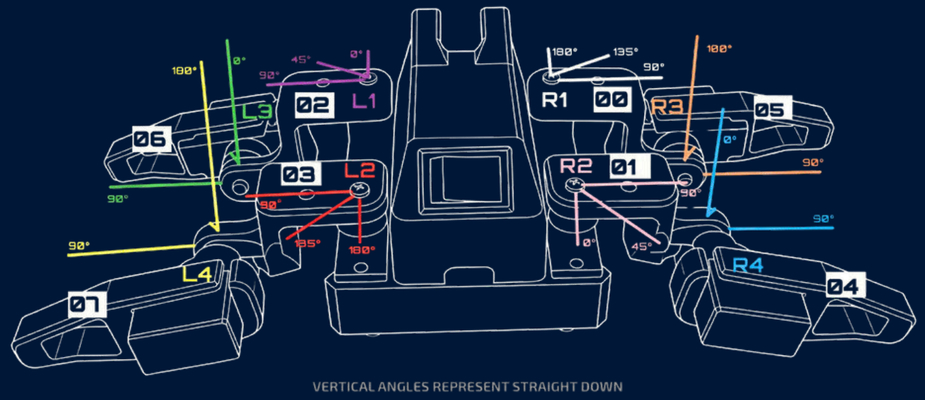
THE QUADRUPED STRUCTURE ALLOWED THE ROBOT TO COMMUNICATE MORE NATURALLY THROUGH POSTURE AND LOCOMOTION COMPARED TO WHEELED ROBOTIC SYSTEMS. QUADRUPED MOVEMENT ALSO CREATES STRONGER EMOTIONAL RELATABILITY BECAUSE ANIMAL-INSPIRED MOTION FEELS MORE FAMILIAR AND ALIVE TO USERS.

THE CAD SYSTEM SUPPORTS MODULAR ASSEMBLY, ACCESSIBLE ELECTRONICS INTEGRATION, STABLE WEIGHT DISTRIBUTION, AND EXPRESSIVE ARTICULATION WHILE REMAINING COMPACT AND FABRICATION-FRIENDLY.

MECHANICAL ASSEMBLY

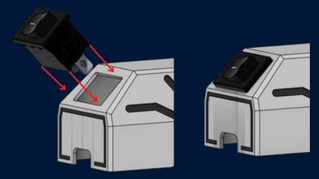
HIP JOINTS



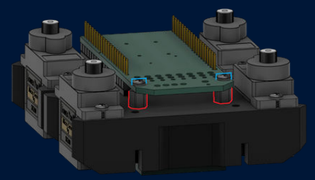


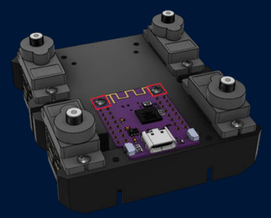
VERTICAL ANGLES REPRESENT STRAIGHT DOWN

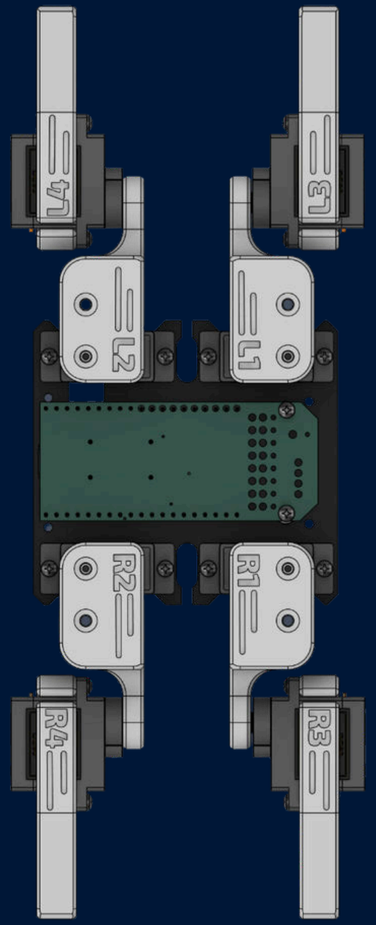
TOP COVER



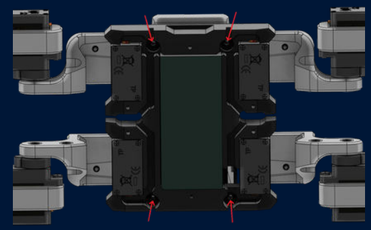
MAIN ELECTRONICS

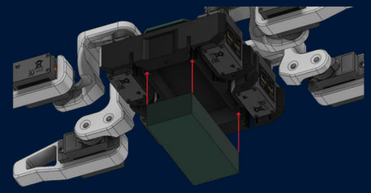


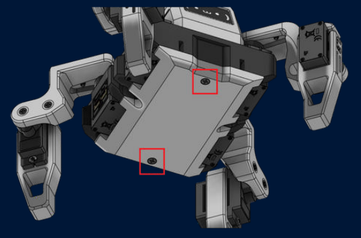




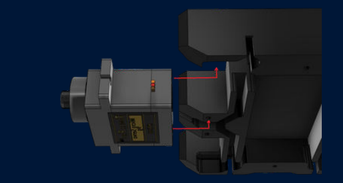
BATTERY ASSEMBLY

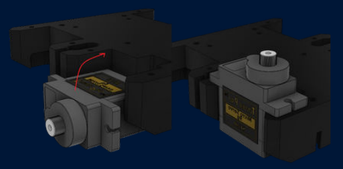


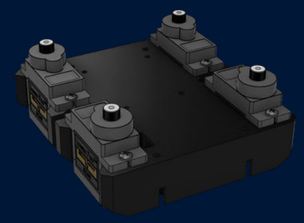




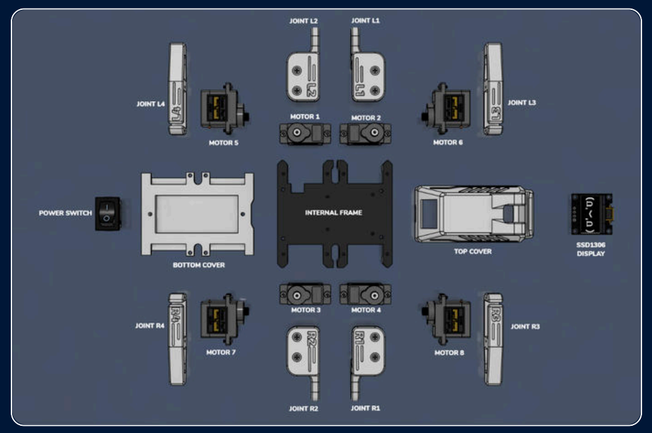
FRAME MOTORS







COMPLETE SET



Technical Architecture

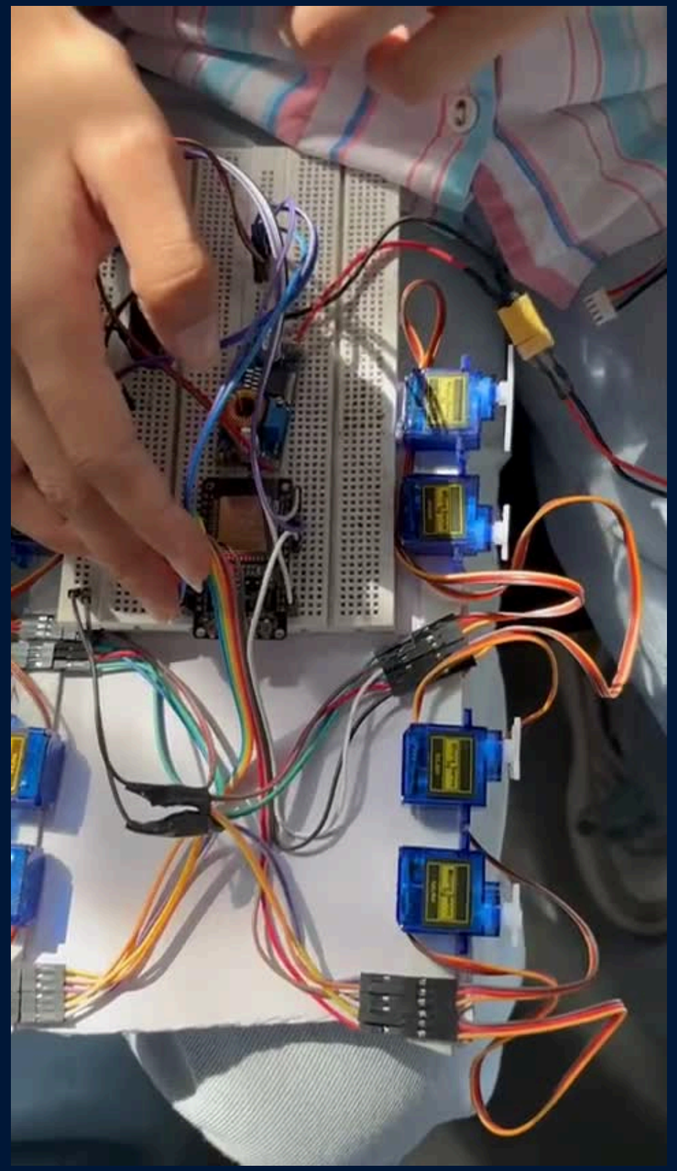
MOSHI IS BUILT AROUND AN ESP32 MICROCONTROLLER SYSTEM INTEGRATED WITH AN 8-SERVO LOCOMOTION SETUP, OLED FACIAL DISPLAY, WIRELESS COMMUNICATION ARCHITECTURE, AND BROWSER-BASED INTERACTION SYSTEM.

THE ESP32 ACTS AS THE CENTRAL PROCESSING UNIT RESPONSIBLE FOR MOVEMENT CONTROL, WIFI COMMUNICATION, EXPRESSION RENDERING, AND INTERACTION MANAGEMENT. THE 8-SERVO ARTICULATION SYSTEM ENABLES EXPRESSIVE QUADRUPED LOCOMOTION AND POSTURE-BASED INTERACTION.

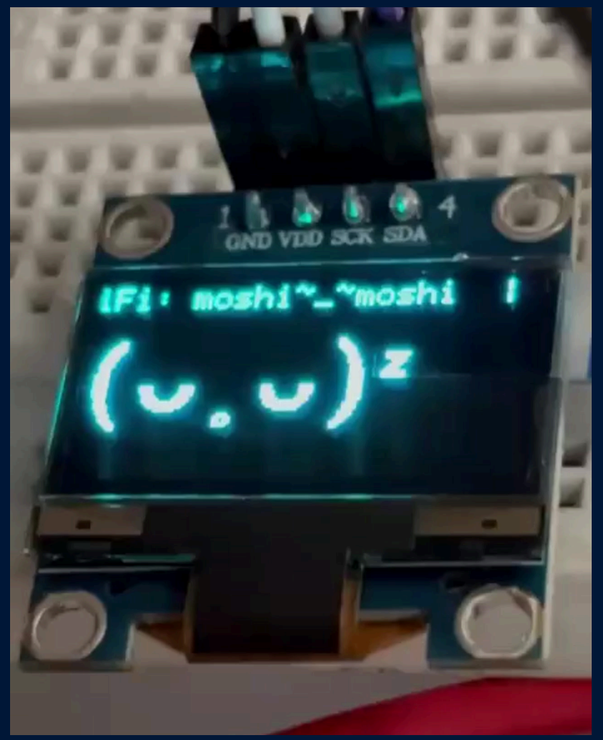
THE OLED DISPLAY FUNCTIONS AS MOSHI'S EMOTIONAL COMMUNICATION LAYER BY RENDERING BLINKING ANIMATIONS, VISUAL EXPRESSIONS, AND SYNCHRONIZED BEHAVIORAL FEEDBACK.
THE SYSTEM ARCHITECTURE COMBINES EMBEDDED SYSTEMS, PHYSICAL COMPUTING, AND INTERACTION DESIGN INTO A SINGLE RESPONSIVE ROBOTIC PLATFORM.







Component breakdown



THE PROJECT INTEGRATES MULTIPLE HARDWARE SYSTEMS THAT WORK TOGETHER TO CREATE EXPRESSIVE ROBOTIC INTERACTION.

THE ESP32 MICROCONTROLLER WAS SELECTED BECAUSE OF ITS INTEGRATED WIFI SUPPORT, COMPACT SIZE, AFFORDABILITY, AND COMPATIBILITY WITH BROWSER-BASED INTERACTION SYSTEMS. THE 8-SERVO ARTICULATION SYSTEM PROVIDES MOVEMENT FLEXIBILITY AND EXPRESSIVE POSTURE CONTROL WHILE MAINTAINING LIGHTWEIGHT MOBILITY.

THE OLED DISPLAY ACTS AS MOSHI'S VISUAL EXPRESSION SYSTEM, ALLOWING THE ROBOT TO COMMUNICATE EMOTION AND RESPONSIVENESS THROUGH BLINKING, EYE MOVEMENT, AND ANIMATED EXPRESSIONS.

THE ENTIRE ROBOT STRUCTURE WAS FABRICATED USING 3D PRINTED PLA COMPONENTS, ENABLING RAPID PROTOTYPING, LIGHTWEIGHT FABRICATION, MODULAR ASSEMBLY, AND ITERATIVE DEVELOPMENT.

Software & Captive Portal

ONE OF THE MOST IMPORTANT PARTS OF THE PROJECT WAS THE DEVELOPMENT OF A CUSTOM CAPTIVE PORTAL INTERACTION SYSTEM.

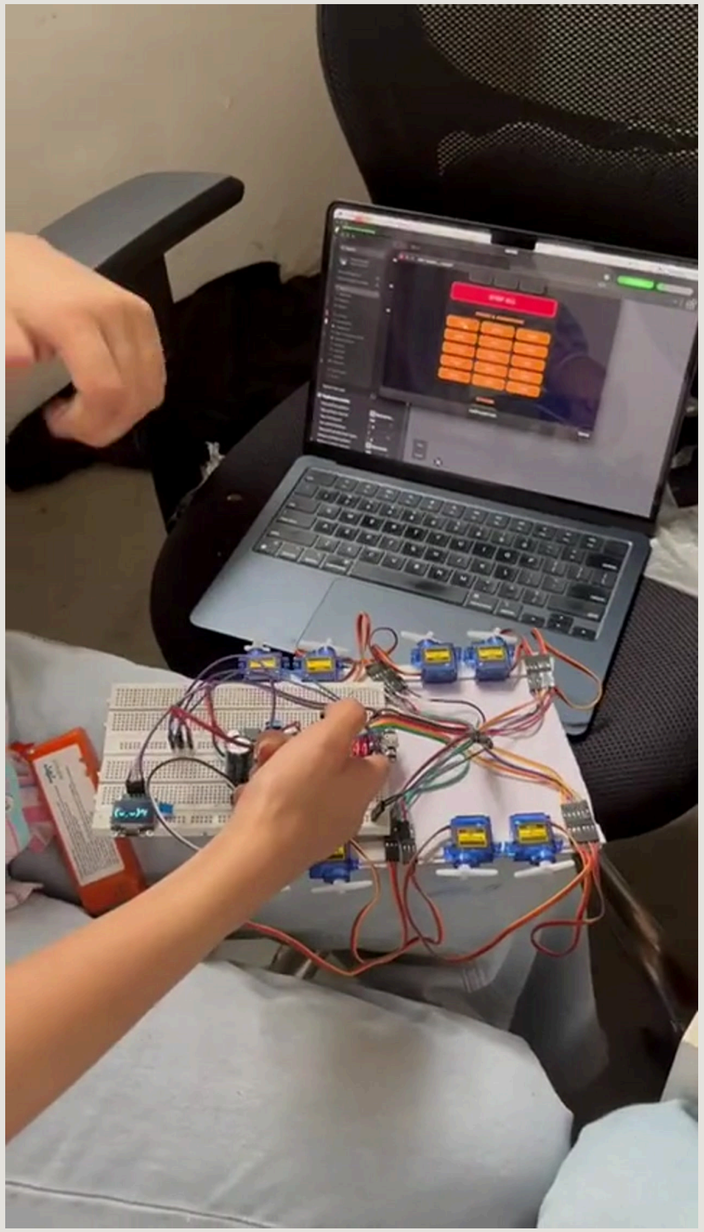
UNLIKE CONVENTIONAL ROBOTICS PROJECTS THAT RELY HEAVILY ON CODING INTERFACES, MOSHI IS PRIMARILY CONTROLLED THROUGH A BROWSER-BASED CAPTIVE PORTAL CONNECTED VIA WIFI. THIS ALLOWS USERS TO CONTROL THE ROBOT USING ANY DEVICE WITH A BROWSER, MAKING THE INTERACTION SYSTEM SIGNIFICANTLY MORE ACCESSIBLE AND USER-FRIENDLY.

THROUGH THE CAPTIVE PORTAL, USERS CAN CONTROL MOSHI'S MOVEMENT IN ALL FOUR DIRECTIONS, TRIGGER BEHAVIORAL ACTIONS, AND SWITCH EMOTIONAL EXPRESSIONS IN REAL TIME. THE INTERACTION SYSTEM TRANSFORMS MOSHI FROM A PURELY TECHNICAL PROTOTYPE INTO A CONNECTED INTERACTIVE PRODUCT.

THE SOFTWARE SYSTEM ALSO INCLUDES MOVEMENT CHOREOGRAPHY, ANIMATION RENDERING, BEHAVIORAL SEQUENCING, WIFI COMMUNICATION, AND SYNCHRONIZED EXPRESSION CONTROL.

THE INTERACTION FLOW FOLLOWS:
USER INPUT → CAPTIVE PORTAL → ESP32 PROCESSING → MOTION + EXPRESSION OUTPUT

THIS INTERACTION MODEL BECAME CENTRAL TO THE PROJECT BECAUSE IT ALLOWED EMOTIONAL INTERACTION TO HAPPEN INTUITIVELY AND WIRELESSLY WITHOUT REQUIRING TECHNICAL PROGRAMMING KNOWLEDGE.



Interaction Design

MOSHI COMMUNICATES PRIMARILY THROUGH MOVEMENT AND BEHAVIORAL TIMING RATHER THAN SPEECH.

THE INTERACTION SYSTEM EXPLORES THE IDEA THAT MOVEMENT ITSELF CAN ACT AS AN INTERFACE. SMALL BEHAVIORAL ACTIONS SUCH AS POSTURE CHANGES, PAUSES, BLINKING, MOVEMENT RHYTHM, AND REACTION TIMING CREATE EMOTIONAL INTERPRETATION AND PERSONALITY.

THE PROJECT DRAWS INSPIRATION FROM ANIMATION PRINCIPLES AND ANIMAL BEHAVIOR STUDIES WHERE SUBTLE MOVEMENT DETAILS SIGNIFICANTLY INFLUENCE EMOTIONAL PERCEPTION.

THE SYNCHRONIZATION BETWEEN FACIAL ANIMATION AND LOCOMOTION STRENGTHENS THE ROBOT’S EMOTIONAL READABILITY AND INTERACTION QUALITY.

Behaviour system

A MAJOR FOCUS OF THE PROJECT WAS DESIGNING BEHAVIOR RATHER THAN SIMPLY CREATING MOVEMENT.

DIFFERENT EMOTIONAL STATES WERE MAPPED THROUGH COMBINATIONS OF POSTURE, MOTION RHYTHM, FACIAL EXPRESSIONS, AND TIMING SYSTEMS. FOR EXAMPLE, CURIOSITY WAS COMMUNICATED THROUGH FORWARD TILTING AND WIDENED EYES, WHILE CALMNESS WAS REPRESENTED USING SLOW BLINKING AND STEADY POSTURE.

EXCITED BEHAVIOR USED BOUNCING MOVEMENT AND ENERGETIC FACIAL ANIMATIONS, WHILE SLEEPY STATES USED SLOWER MOTION AND DROOPING EYES.
THIS TRANSFORMED LOCOMOTION INTO EMOTIONAL COMMUNICATION AND ALLOWED MOSHI TO BEHAVE MORE LIKE A COMPANION CHARACTER THAN A TRADITIONAL ROBOT.

Features

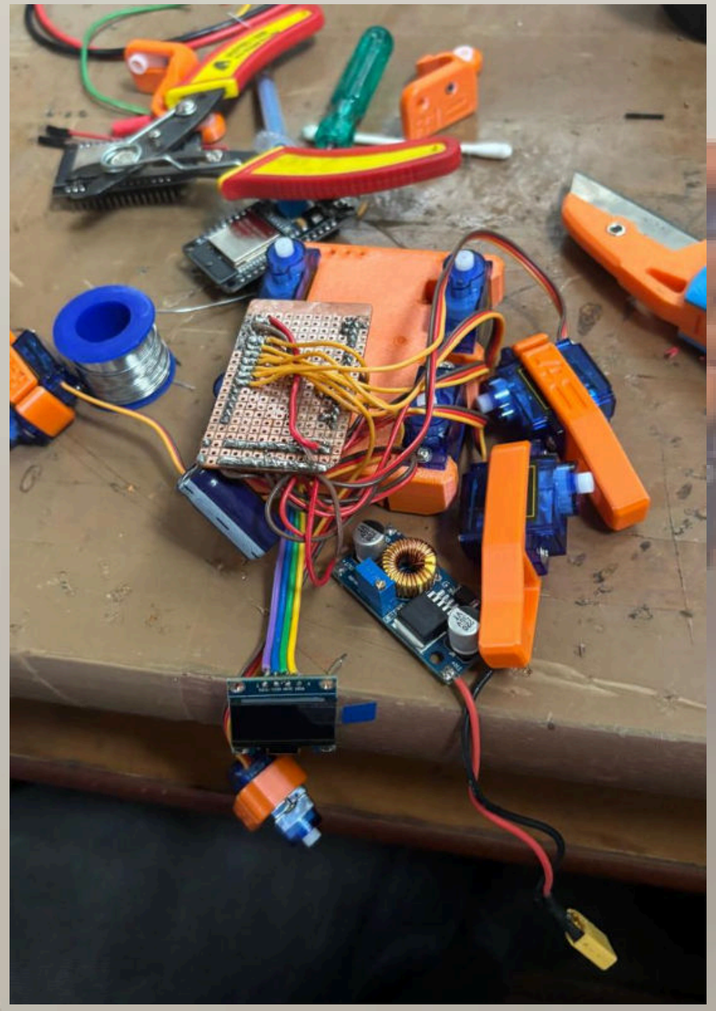
MOSHI COMBINES BOTH FUNCTIONAL AND EXPERIENTIAL FEATURES.

FROM A TECHNICAL PERSPECTIVE, THE ROBOT INCLUDES QUADRUPED LOCOMOTION, WIRELESS BROWSER-BASED INTERACTION, EXPRESSIVE OLED FACIAL SYSTEMS, MOTION SEQUENCING, API INTERACTION SYSTEMS, AND BEHAVIORAL ANIMATION CONTROL.

FROM AN EXPERIENTIAL PERSPECTIVE, THE ROBOT FOCUSES ON EMOTIONAL INTERACTION, PLAYFUL MOVEMENT, IDLE ANIMATIONS, CURIOSITY-DRIVEN BEHAVIOR, EXPRESSIVE BLINKING, AND APPROACHABLE COMMUNICATION SYSTEMS.

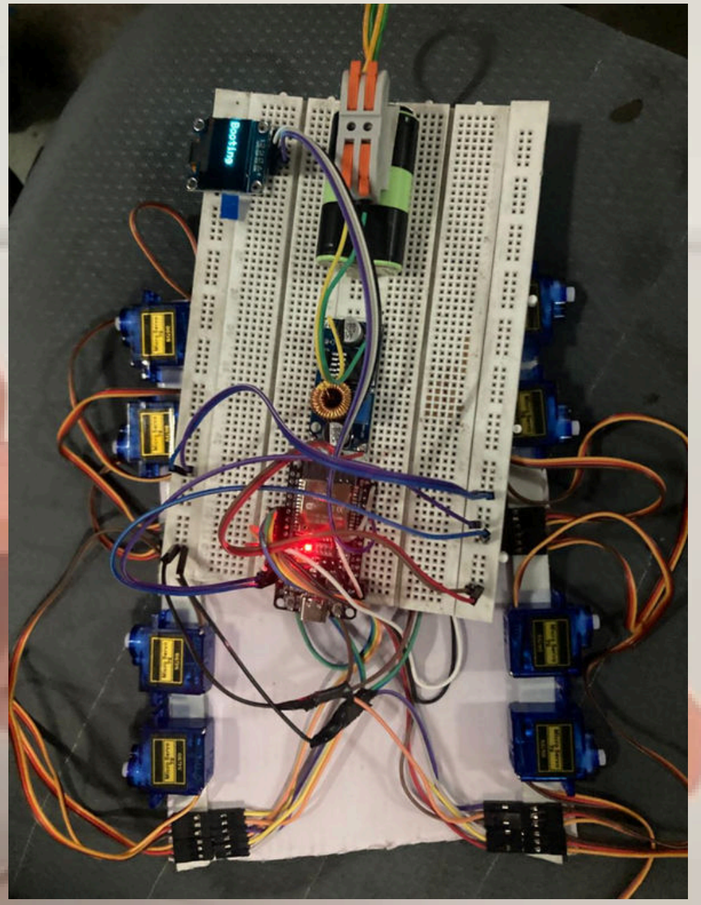
THE COMBINATION OF TECHNICAL FUNCTIONALITY AND EMOTIONAL DESIGN CREATES A MORE ENGAGING INTERACTION EXPERIENCE.

Challenges Faced









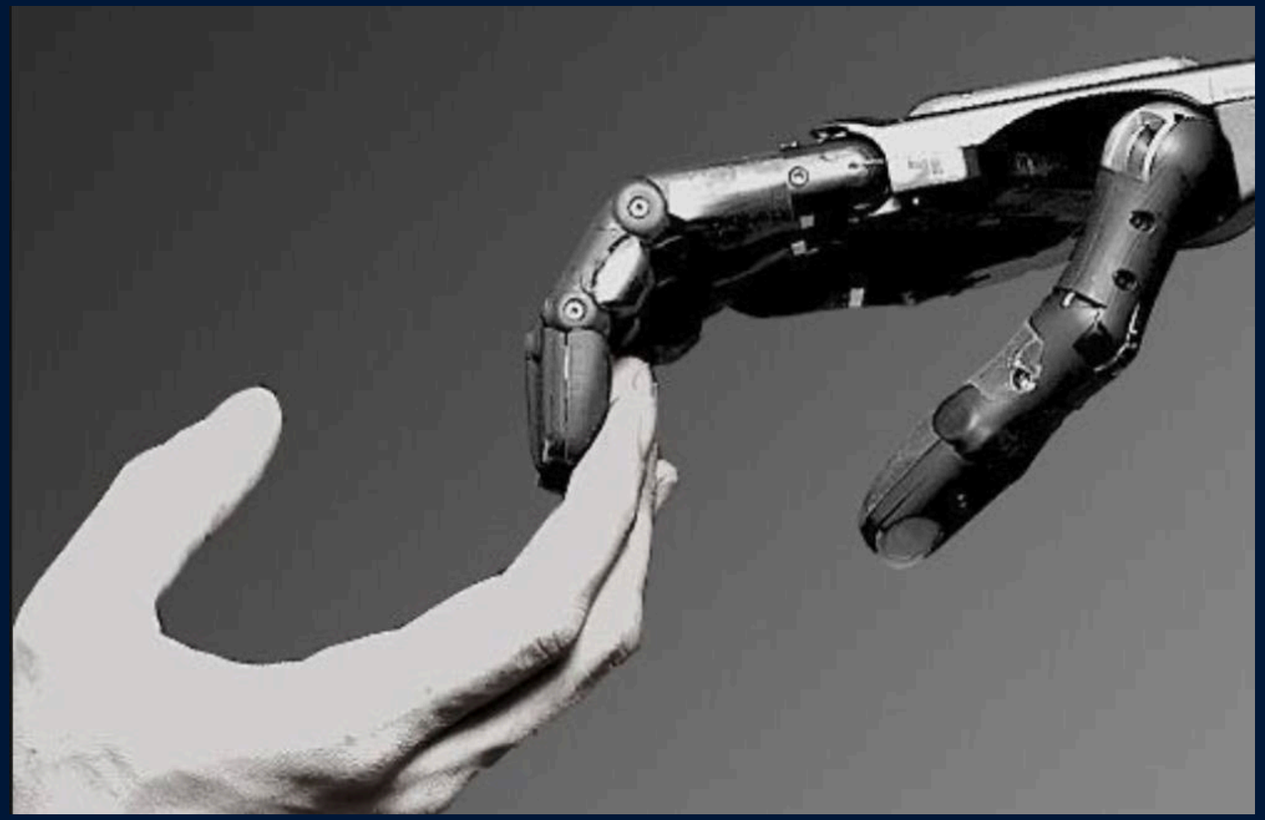
Iterations made

User experience Scenario

MOSHI WAS IMAGINED AS A DESKTOP COMPANION FOR STUDENTS, DESIGNERS, AND CREATIVES WORKING LONG HOURS IN ISOLATED OR DIGITALLY INTENSIVE ENVIRONMENTS.

RATHER THAN FUNCTIONING PURELY AS A UTILITY ROBOT, MOSHI CREATES MOMENTS OF CURIOSITY, EMOTIONAL ENGAGEMENT, PLAYFUL DISTRACTION, AND INTERACTIVE COMPANIONSHIP THROUGH EXPRESSIVE MOVEMENT AND RESPONSIVE BEHAVIOR.

THE PROJECT EXPLORES HOW EMOTIONALLY EXPRESSIVE ROBOTICS COULD EVENTUALLY BECOME PART OF EVERYDAY ENVIRONMENTS AS INTERACTIVE COMPANIONS AND BEHAVIORAL INTERFACES.



Future Oppurtunities

THE PROJECT OPENS POSSIBILITIES FOR FUTURE DEVELOPMENTS INCLUDING AI-ASSISTED INTERACTION, VOICE COMMUNICATION SYSTEMS, TOUCH-SENSITIVE FEEDBACK, EMOTION RECOGNITION, ADAPTIVE BEHAVIORAL SYSTEMS, AUTONOMOUS NAVIGATION, AND COMPUTER VISION INTEGRATION.

MOSHI ALSO CREATES OPPORTUNITIES WITHIN EDUCATIONAL ROBOTICS, THERAPEUTIC INTERACTION SYSTEMS, EMOTIONAL AI RESEARCH, INTERACTIVE INSTALLATIONS, AND SOCIAL ROBOTICS APPLICATIONS.

THE PROJECT DEMONSTRATES HOW OPEN-SOURCE ROBOTICS PLATFORMS CAN EVOLVE INTO EMOTIONALLY INTELLIGENT INTERACTION SYSTEMS THROUGH DESIGN INTERVENTION.

Learnings

ONE OF THE BIGGEST LEARNINGS FROM THE PROJECT WAS UNDERSTANDING THAT MOVEMENT ITSELF CAN COMMUNICATE EMOTION AND PERSONALITY.

THE PROJECT DEMONSTRATED THAT SMALL BEHAVIORAL INTERACTIONS SUCH AS BLINKING, PAUSES, POSTURE SHIFTS, AND MOVEMENT TIMING SIGNIFICANTLY INFLUENCE EMOTIONAL ATTACHMENT AND USER PERCEPTION.

IT ALSO HIGHLIGHTED HOW INTERACTION DESIGN CAN HUMANIZE TECHNOLOGY AND MAKE ROBOTICS FEEL MORE APPROACHABLE, ACCESSIBLE, AND EMOTIONALLY ENGAGING.

THE DEVELOPMENT PROCESS EMPHASIZED THAT BEHAVIORAL SYSTEMS AND EMOTIONAL COMMUNICATION ARE JUST AS IMPORTANT AS TECHNICAL FUNCTIONALITY WITHIN INTERACTIVE ROBOTICS.

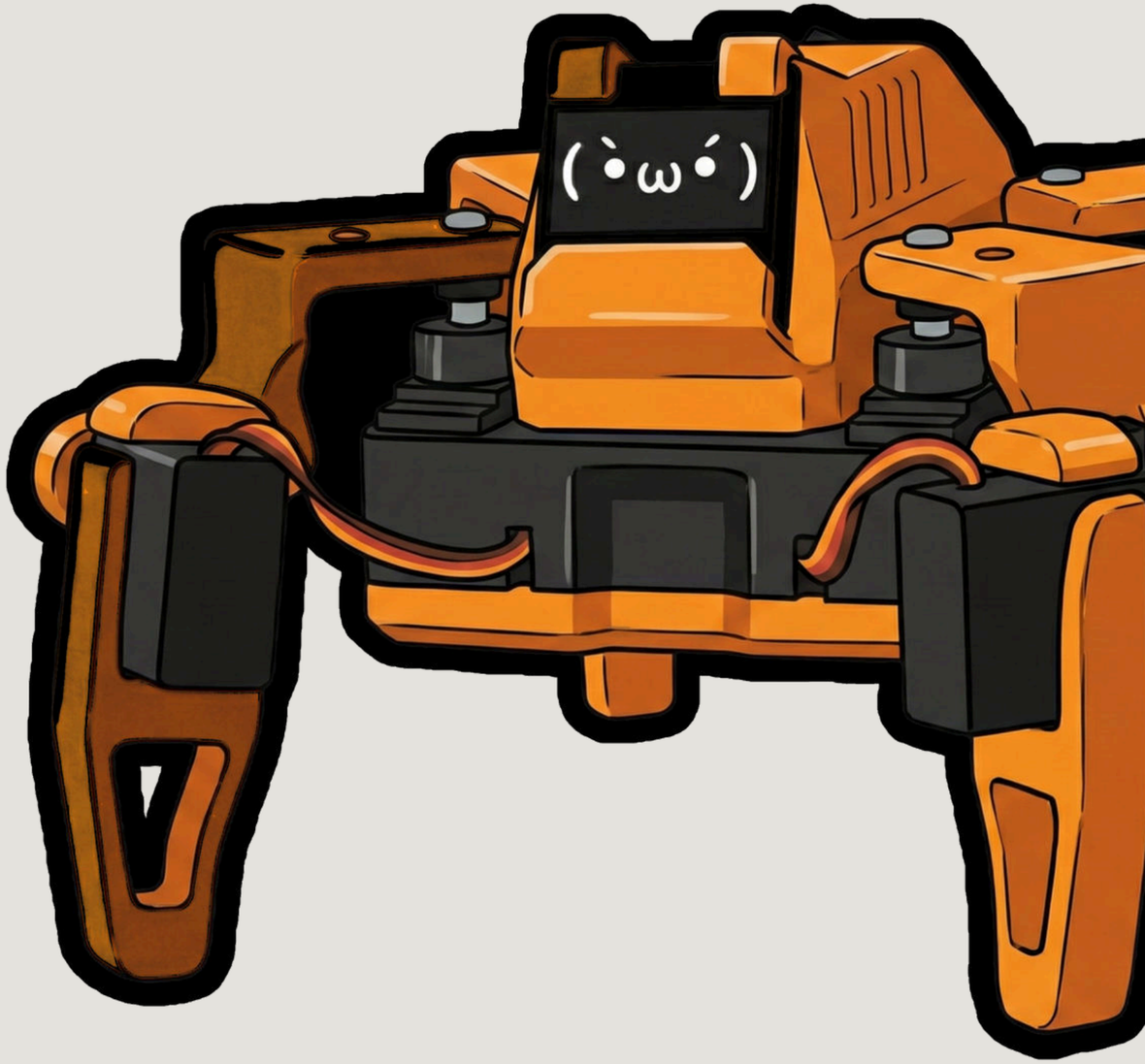
Conclusion

MOSHI DEMONSTRATES HOW ROBOTICS CAN EVOLVE BEYOND PURELY FUNCTIONAL ENGINEERING SYSTEMS INTO EMOTIONALLY EXPRESSIVE INTERACTIVE COMPANIONS.

THE PROJECT COMBINES INTERACTION DESIGN, EMOTIONAL COMPUTING, ROBOTICS, PHYSICAL AI, BEHAVIORAL ANIMATION, AND EMBEDDED SYSTEMS INTO A SINGLE HUMAN-CENTERED INTERACTIVE EXPERIENCE.

THROUGH EXPRESSIVE MOVEMENT, WIRELESS INTERACTION, BEHAVIORAL TIMING, AND EMOTIONAL COMMUNICATION, MOSHI EXPLORES A FUTURE WHERE ROBOTICS BECOMES EMOTIONALLY INTEGRATED INTO EVERYDAY LIFE RATHER THAN REMAINING PURELY MECHANICAL TECHNOLOGY.

THE PROJECT ULTIMATELY REPRESENTS THE INTERSECTION OF DESIGN, TECHNOLOGY, AND EMOTIONAL INTERACTION WITHIN EMERGING COMPANION ROBOTICS SYSTEMS.



MECHANICAL ASSEMBLY

CAN ROBOTICS FEEL EMOTIONALLY APPROACHABLE?

Moshi is an expressive robotic companion exploring how movement, behavior, and interaction can create emotional connection between humans and machines.

MOSHI EXPLORES A FUTURE WHERE ROBOTICS BECOME EMOTIONALLY ENGAGING EVERYDAY COMPANIONS.

Moshi is a quadruped companion robot that combines embedded systems, expressive interaction, and wireless control within a compact modular platform. Built using an ESP32 microcontroller, servo-based locomotion, and an OLED facial display, Moshi uses a custom captive portal interface to enable real-time emotional expression and directional movement through a browser-based interaction system.

ELECTRONICS / WIRING

Moshi's electronics system is built around an ESP32 microcontroller that controls movement, emotional behaviors, WiFi connectivity, and real-time interaction. The robot uses an 8-servo articulation system for expressive quadruped motion and an OLED display for animated facial expressions. A custom captive portal interface allows wireless control of emotions and directional movement through a browser-based dashboard.

CURRENT MARKET OPPORTUNITIES OF MOSHI

The growing interest in companion robotics, emotional AI, and interactive consumer technology creates strong opportunities for systems like Moshi. With increasing demand for educational robots, therapeutic companions, desk assistants, and emotionally engaging AI products, Moshi explores how expressive robotics can evolve beyond functionality into approachable, human-centered everyday experiences.

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THANK YOU!

MOSHI

Exploring Emotional Interaction Through Companion Robotics

Ilisha Dabas
2k24/BD/046